SOFTWARE TESTING

Module 1(Assignments)

1. What is SDLC

SDLC stands for software development life cycle that provide model for development.

There are 6 phases

- Requirement Gathering and Analysis
- Designing
- Implementation
- Testing
- Deployment
- Maintenance

2. What is software testing?

Software testing is a process used to identify the correctness, completeness, and quality of developed computer software.

3.What is agile methodology?

Agile means delivering working software in small parts with continuous feedback and improvements.

Why Agile:

1. Quick Delivery
2. Easy to change to requirement
3. Continuous collaboration with customers **Agile Key point:**

1. Iterative Incremental model
2. Work divided into small parts
3. Need to complete work at a specific time
4. Daily meeting
5. Continuous improvement
6. Continuous testing and feedback
7. Customers are involved in every phase
8. Fast and flexible delivery

4. What is SRS

A software requirements specification (SRS) is a complete description of the behaviour of the system to be developed.

It includes a set of use cases that describe all of the interactions that the users will have with the software.

5. What is oops

OOPs: Object Oriented Programming To way to write code Less code Redundancy Reusability Secure

6. Write Basic Concepts of oops

Concepts of OOPS:

Class

Object

Inheritance

Encapsulation

Polymorphism

Abstraction

7. What is object

Object gives permission to access functionality of class

8. What is class

It is a collection of data member and member functions

9. What is encapsulation

Encapsulation is a Wrapping of data

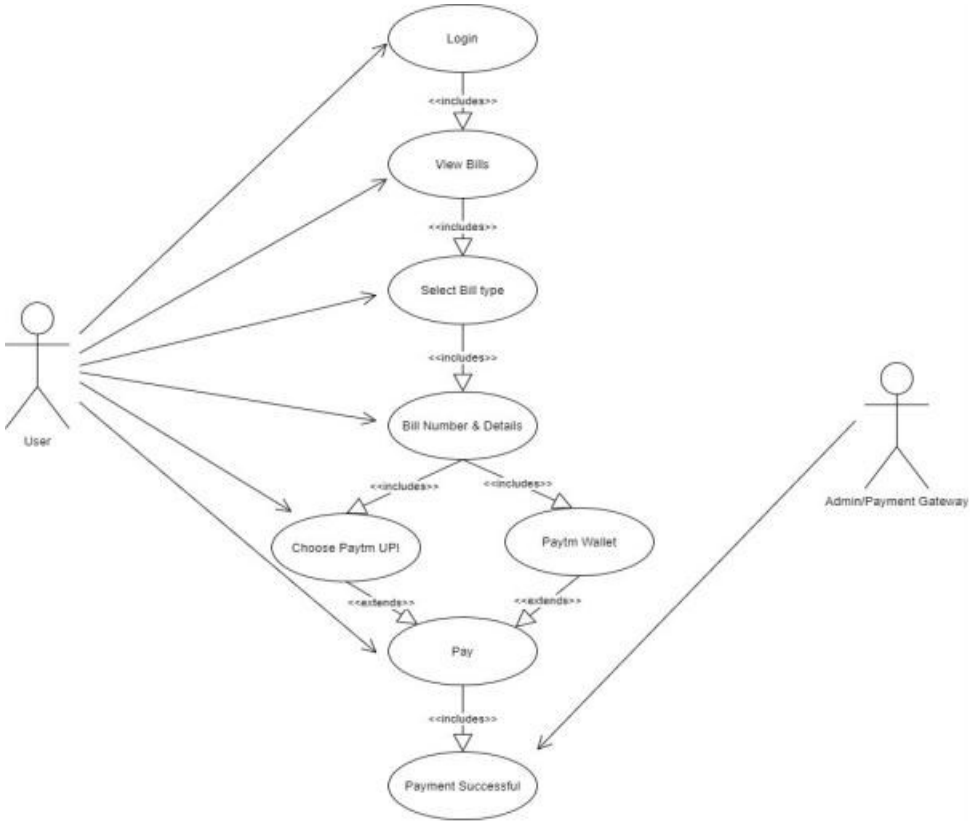
10. What is inheritance

To derive attribute of some other class

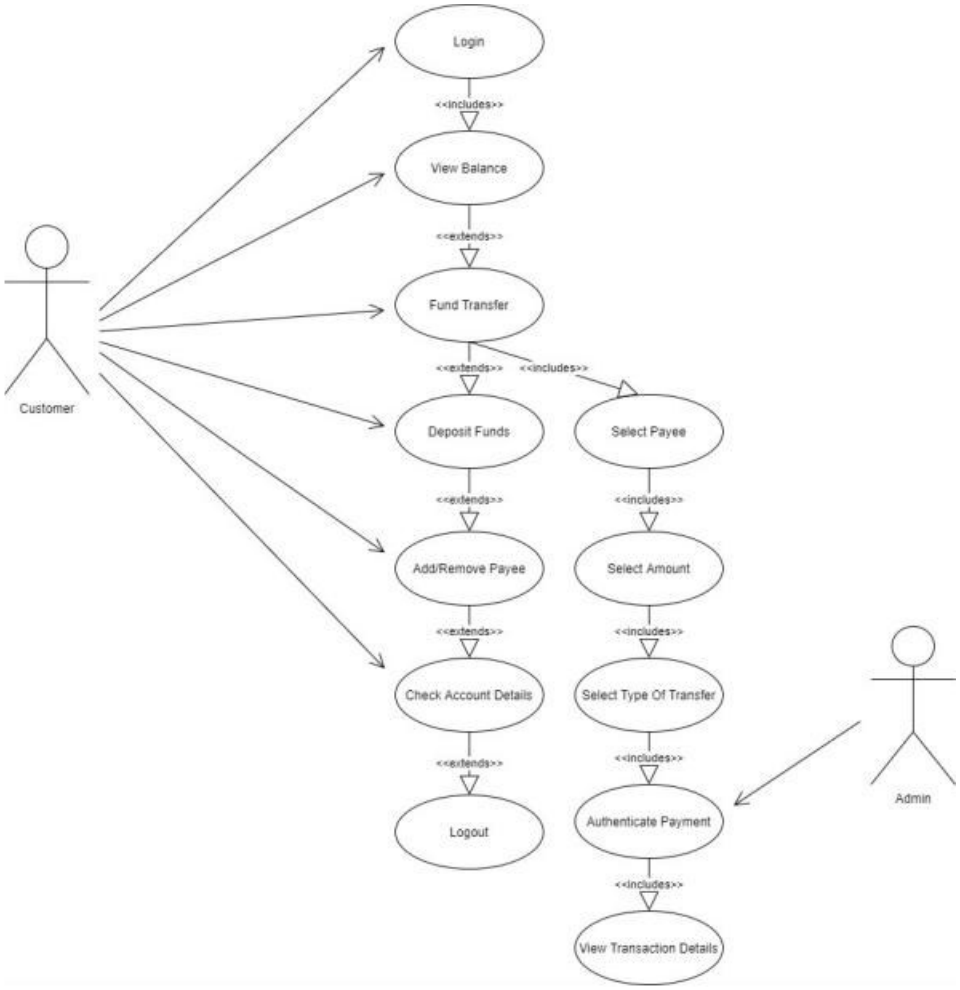
11. What is polymorphism

The meaning of polymorphism one name multiple form

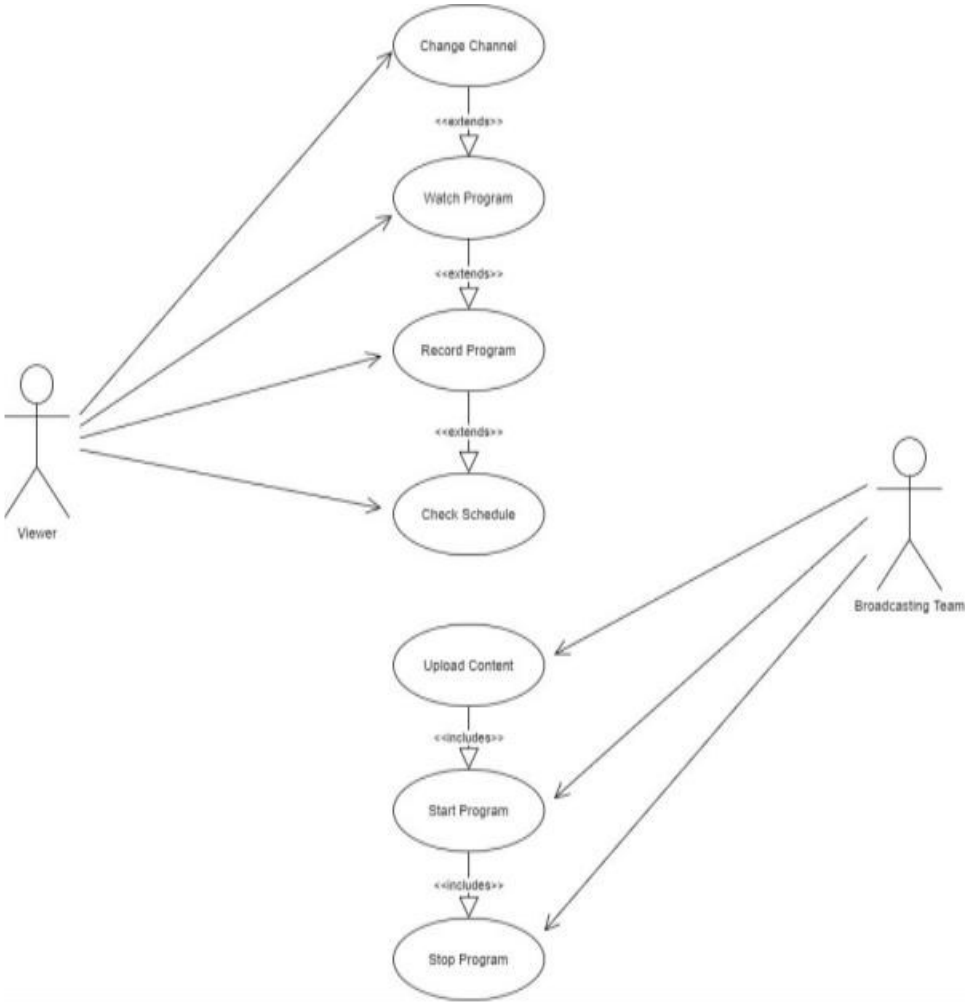
12. Draw Usecase on online bill payment system (paytm)



13.Draw Usecase on banking system for customers



14.Draw Usecase on Broadcasting System



15. Write SDLC phases with basic introduction

1. Requirement Gathering:

- Collect Details from client about the project like
  Functional Requirements and Non- Functional Requirements

1.features	1.performance
2.UI requirements	2.security
3. functionality of the feature	3.usability

2. Analysis: To Analysis the requirements and convert the CRS into SRS
3. Design: To prepare Design/UI as per the requirements
4. Implementation: Coding
5. Testing and Deployment: To verify the correctness, completeness and quality of the software developed and deploy it
6. Maintenance:
   1.corrective: to repair a defect
   2.adaptive: to test in new platform
   3.perfective: to add a new feature

16. Explain Phases of the waterfall model

The classical software lifecycle models the software development is a step- by-step "waterfall" between the various development phases.

1. Requirement Analysis: Gather and analysis all project requirements. Understand what the customer needs. Document requirements in an SRS (Software Requirement Specification).
2. System Design: Design system architecture. Create database design and system flow. prepare design documents.
3. Implementation: Developers write code according to design. each module is developed separately.
4. Testing: Test the developed software. Identify and fix bugs. Perform unit
5. Deployment: Install the software in the customer environment. make the system available for users.
6. Maintenance: Fix errors found after deployment. Update and improve the system. Add new features if required.

17. Write phases of spiral model

The **Spiral Model** is a software development process that combines features of both **Waterfall** and **Prototyping** models. It focuses mainly on **risk analysis** and is used for large, complex, and high-risk projects.

1. planning phase:

- Determination of objectives, alternatives and constraints
- Initial requirements

2. Risk Analysis phase:

- Analysis of alternatives and identification/resolution of risk
- Something that will delay project or increase its cost

3. Engineering:

- Development of the "next level" product
- Evolving system

4. Customer Evolution:

- Assessment of the result of engineering
- Alpha demo
- Completion

18. Write agile manifesto principles

The **Agile Manifesto** was created in 2001 by 17 software developers at Snowbird Resort. It defines **12 principles** that guide Agile software development.

12 Principles of Agile Manifesto

1. **Customer Satisfaction Through Early and Continuous Delivery** Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.
2. **Welcome Changing Requirements**
   Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.
3. **Frequent Delivery**
   Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to shorter timescales.
4. **Business and Developers Collaboration**
   Business people and developers must work together daily throughout the project.
5. **Build Projects Around Motivated Individuals**
   Give them the environment and support they need, and trust them to get the job done.
6. **Face-to-Face Communication**
   The most efficient and effective method of conveying information is face-toface conversation.
7. **Working Software is the Primary Measure of Progress** Working software is the main measure of progress.
8. **Sustainable Development**
   Agile processes promote sustainable development. Sponsors, developers, and users should be able to maintain a constant pace indefinitely.
9. **Technical Excellence and Good Design**
   Continuous attention to technical excellence and good design enhances agility.
10. **Simplicity**
    Simplicity—the art of maximizing the amount of work not done—is essential.
11. **Self-Organizing Teams**
    The best architectures, requirements, and designs emerge from self-organizing teams.
12. **Regular Reflection and Improvement**
    At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behaviour accordingly.

19.Explain working methodology of agile model and also write pros and cons.

The working methodology of agile model is the project scope and creating a prioritized product backlog of features.
The sprint planning teams select items from the backlog to complete during the upcoming, time-boxed sprint.
Development and testing Cross-functional teams work concurrently on designing, coding, and testing features to ensure quality

Daily stands-up daily meetings to discuss progress, synchronize efforts, and address obstacle
At the end of the cycle, the team demonstrates the product to stakeholders for feedback and reflects on improving processes.
Continuous improvement the process repeats with the feedback from each iteration informing the next, ensuring the final product meets customer needs.

Pros:
Simple and easy to understand and use

Easy to manage due to the rigidity of the model. Each phase has specific deliverables and a review process.

Phases are processed and completed one at a time.

Works well for smaller projects where requirements are very well understood.

Clearly defined stages.

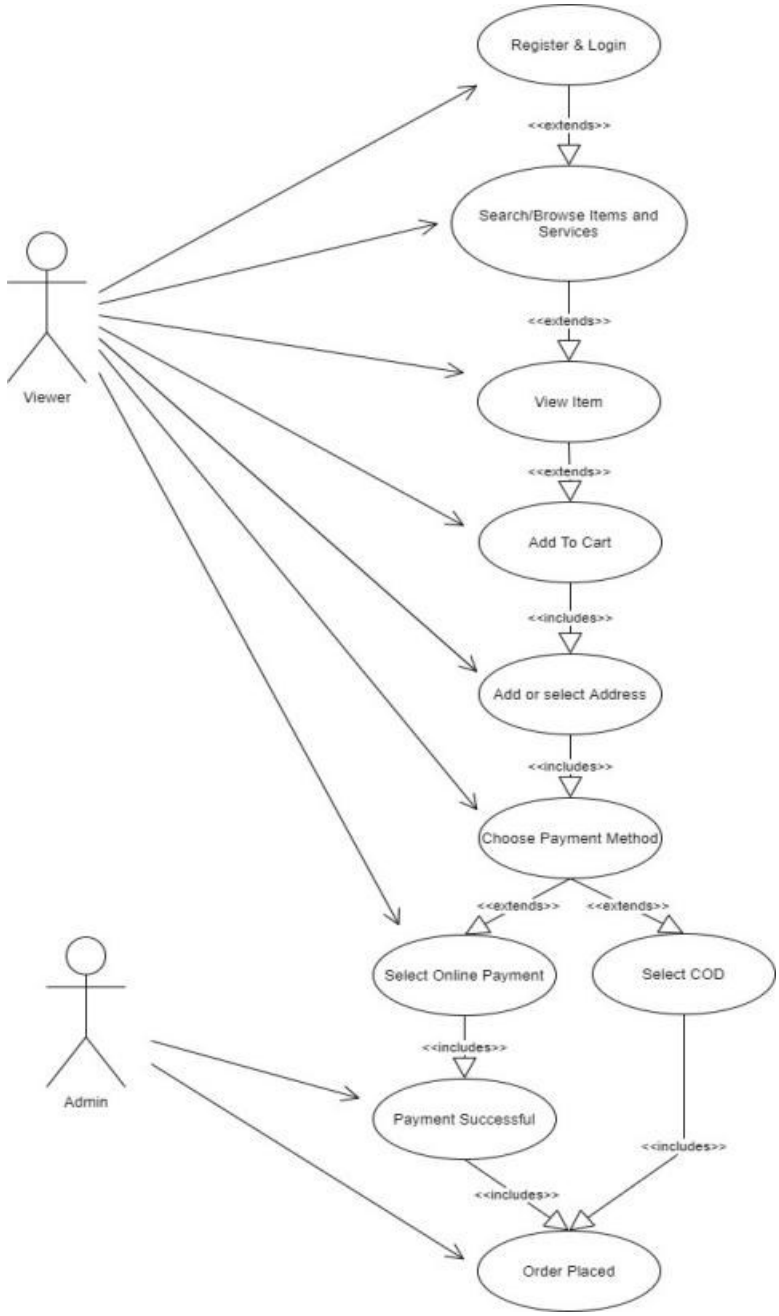
Well-understood milestones. Easy to arrange tasks.

Process and results are well documented

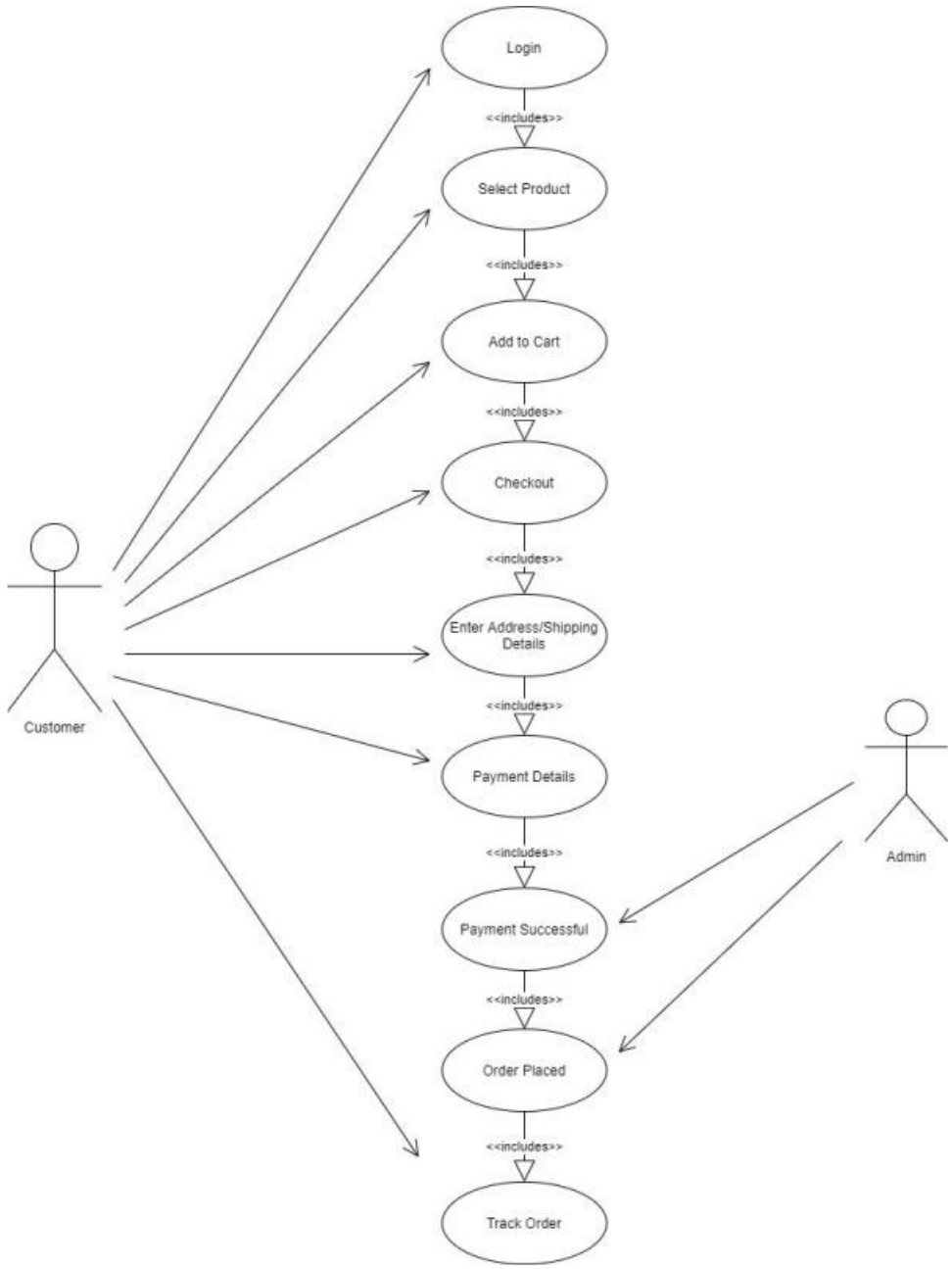
Cons:
No working software is produced until late during the life cycle. High amounts of risk and uncertainty.
Not a good model for complex and object-oriented projects. Poor model for long and ongoing projects.
Not suitable for projects with a moderate to high risk of changing requirements. So risk and uncertainty are high with this process model.
It is difficult to measure progress within stages. Cannot accommodate changing requirements.
No working software is produced until late in the life cycle. Adjusting scope during the life cycle can end a project.
Integration is done as a "big bang. at the very end, which doesn't allow identifying any technological or business bottlenecks or challenges early.

20. Draw usecase on OTT Platform.

21. Draw usecase on E-commerce application



22. Draw usecase on Online shopping product using payment gateway.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